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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The global agricultural sector faces the unprecedented challenge of increasing food production to sustain a human population projected to approach 10 billion by 2050 (Dayioglu & Turker, 2021). To meet these escalating demands while mitigating the adverse impacts of climate change, resource depletion, and environmental degradation, traditional farming methodologies are rapidly shifting towards “Agriculture 4.0” (Abbasi, Martinez, & Ahmad, 2022). This paradigm integrates modern information and communication technologies (ICT), the Internet of Things (IoT), artificial intelligence (AI), and data analytics to optimise agricultural systems (Abiri, Rizan, Balasundram, Shahbazi, & Abdul-Hamid, 2023). At the core of this digital transformation is the necessity for accurate, real-time data collection and on-farm recordkeeping, which serve as the foundation for precision agriculture and operational efficiency (Basir, Buckmaster, Raturi, & Zhang, 2024).

Historically, agricultural practitioners have relied on unstructured, paper-based logbooks or human memory to track operations. While these traditional methods possess inherent provenance, they are highly susceptible to data fragmentation, loss, recall bias, and significant transcription errors (Basir et al., 2024). In response, Farm Management Information Systems (FMIS) evolved from simple digitalised field books into complex platforms designed to collect, process, store, and disseminate data to support human decision-making in farm enterprises (Fountas et al., 2015; Tummers, Kassahun, & Tekinerdogan, 2019). Modern digital recordkeeping platforms streamline administrative burdens, facilitate data interoperability, and enable precise financial tracking and bioprocess monitoring. Empirical adoption studies further establish that uptake of these systems is strongly conditioned by farmers’ education, access to advisory services, and access to rural credit—factors that are unevenly distributed in developing economies (Carrer, de Souza Filho, & Batalha, 2017).

However, the distribution and adoption of these digital agricultural technologies remain highly unequal globally (Javaid, Haleem, Singh, & Suman, 2022; Mhlanga, 2023). In Sub-Saharan Africa, including Nigeria, the integration of advanced FMIS

faces unique socio-economic and infrastructural barriers. The Nigerian agricultural landscape predominantly comprises smallholder farmers and medium-scale enterprises that frequently lack access to robust technological infrastructure and capital (Nirosha, 2024). Recent market analyses by the Global System for Mobile Communications Association (GSMA) reveal that 68% of Nigerians in rural areas lacked smartphones in 2022, with rural ownership stagnating at 32% compared to 58% in urban areas (Jaiyeola, 2023). This severe digital divide is primarily exacerbated by high inflation, device-affordability constraints, and pervasive digital illiteracy (Jaiyeola, 2023; Mhlanga, 2023). Consequently, while the existing 32% smartphone penetration provides a critical leapfrog opportunity to deploy mobile-based agricultural advisory services, comprehensive digital farm record management remains heavily underutilised. Engineers and software developers must therefore contextualise global digital-farming advancements into localised, accessible platforms. Developing tools that merge farm mechanisation tracking, bioprocess monitoring, and financial management into a single decision-support architecture holds the potential to revolutionise agricultural productivity in Nigeria.

1.2 Statement of the Problem

Despite the well-documented benefits of digital agriculture, practitioners and engineers in Nigeria face severe bottlenecks in adopting existing Farm Management Information Systems (FMIS) (Tummers et al., 2019). Current commercial FMIS platforms and Enterprise Resource Planning (ERP) systems frequently exhibit high complexity, lack standardised data formats, and impose prohibitive costs that exclude small and medium-scale farmers (Fountas et al., 2015; Giua, Materia, & Camanzi, 2020).

Furthermore, agricultural operations suffer from critical data fragmentation. Farm managers typically maintain Farm Financial Accounts (FFA) entirely separate from their operational and bioprocess records, resulting in duplicated data entry, administrative fatigue, and analytical inefficiencies (Poppe, Vrolijk, & Bosloper, 2023). Because essential metrics—such as equipment maintenance schedules, labour inputs, crop-yield data, and financial expenditures—are siloed, farmers cannot accurately correlate mechanisation inputs or biological variables with actual financial profitability (Gebresenbet et al., 2023).

Without interoperable, structured data, Nigerian agricultural enterprises cannot deploy predictive analytics or decision-support algorithms to optimise resource utilisation (van Klompenburg, Kassahun, & Catal, 2020). In rural contexts where poor internet connectivity and limited technical training exacerbate these issues, deploying resource-intensive, cloud-heavy ERP solutions proves impractical (Abbasi et al., 2022; Mhlanga, 2023; Nirosha, 2024). Therefore, a critical gap exists for a localised, unified digital platform—AGRI-PROFIT—that bridges financial tracking and operational farm data to generate actionable decision support tailored to the specific constraints of Nigerian agriculture.

1.3 Aim and Objectives

Aim

The primary aim of this research is to design, develop, and evaluate AGRI-PROFIT, a secure, integrated farm record management and decision-support platform that overcomes socio-technical adoption barriers to optimise resource allocation, mechanisation tracking, and stakeholder data-sharing for Nigerian agricultural enterprises.

Objectives

To achieve this overarching aim, the specific, measurable objectives of the study are:

1. To identify and evaluate the socio-technical barriers—specifically data fragmentation, interoperability constraints, and user complexity—that systematically hinder the widespread adoption of digital Farm Management Information Systems (FMIS) in the Nigerian context (Carrer et al., 2017; Tummers et al., 2019).
2. To formulate a conceptual system architecture for the AGRI-PROFIT platform that seamlessly translates static operational and financial farm records into dynamic, actionable decision-support metrics for optimising farm mechanisation and input allocation (Gebresenbet et al., 2023).
3. To design and integrate a secure data-sharing framework within the platform architecture that standardises yield and profit/loss reporting, serving as

a verifiable analytical tool to build operational trust among agricultural banks, investors, and policymakers (Huang, 2020; Abiri et al., 2023).

4. To implement and empirically evaluate the AGRI-PROFIT platform's usability, cost-effectiveness, and operational viability within the specific infrastructural constraints of Nigerian smallholder and medium-scale farms.

1.4 Justification of the Study

The development of the AGRI-PROFIT platform delivers immediate economic and operational benefits to agricultural producers while significantly advancing the discipline of Agricultural and Bio-resources Engineering. For farmers and agribusiness managers, the system addresses the critical need to mitigate administrative burdens and overcome data fragmentation (Poppe et al., 2023). By synthesising operational field records with financial data, AGRI-PROFIT empowers farm managers to transition from intuition-based strategies (Husemann & Novkovic, 2012) to evidence-based precision farming. This transition allows producers to precisely calculate production costs, reduce resource wastage, and maximise gross margins at the specific crop or livestock level (Basir et al., 2024).

Furthermore, this research specifically addresses the “human element” of digital agriculture by targeting the socio-technical barriers—such as poor software interoperability and user complexity—that cause existing systems to fail (Carrer et al., 2017; Tummers et al., 2019). By formulating a robust conceptual system architecture, the platform addresses a major macro-economic problem for the agricultural sector: the lack of credible, standardised data required by agricultural banks and investors to approve financing or insurance (Gebresenbet et al., 2023; Huang, 2020).

For Agricultural and Bio-resources Engineers, this research demonstrates the vital intersection of cyber-physical systems, data analytics, and agricultural management, often termed AgInformatics (Dayioglu & Turker, 2021). As the Nigerian agricultural sector seeks to modernise, engineers must move beyond designing physical machinery to developing the software infrastructures required to manage those machines efficiently. Developing localised digital tools ensures that Nigeria can sustainably participate in the Agriculture 4.0 revolution, ultimately driving long-term food security and economic resilience (Mhlanga, 2023).

1.5 Scope and Limitations of the Study

Scope

This research encompasses the software-engineering lifecycle—including system architecture design, coding, and initial deployment—of the AGRI-PROFIT platform, tailored specifically for Nigerian agricultural practices. The platform primarily targets core FMIS functionalities: digital operational logbooks, financial-accounting integration, mechanisation tracking, and decision support for standard crop and livestock enterprises.

Limitations

The study acknowledges specific constraints inherent to the operating environment. Given the high capital costs and limited rural network infrastructure in Nigeria, AGRI-PROFIT prioritises manual digital inputs via smartphones or computers rather than relying on fully automated, capital-intensive IoT sensor networks (Abbasi et al., 2022; Basir et al., 2024). In keeping with these realities, offline capability is implemented primarily for the data-entry (write) path, so that farmers can record operations and transactions without connectivity; data retrieval and synchronisation then reconcile automatically once a connection becomes available (Mhlanga, 2023; Nirosha, 2024). Furthermore, while immutable distributed-ledger (blockchain) governance is reviewed in Section 2.5 as a mechanism for stakeholder trust, it is treated in this study as a conceptual design direction and future extension rather than part of the current implementation; within the present scope, secure data-sharing is achieved through authenticated, access-controlled, tokenised standardised reporting. The widespread deployment of the platform also faces infrastructural headwinds; notably, the 68% rural smartphone-usage gap (Jaiyeola, 2023) means the initial rollout will inherently target early-adopting smallholders and medium-scale enterprises with existing mobile access. Finally, the decision-support modules employ deterministic, rule-based modelling complemented by lightweight statistical regression rather than advanced deep-learning algorithms, ensuring the computational requirements remain feasible for accessible mobile processors (van Klompenburg et al., 2020).

CHAPTER TWO

LITERATURE REVIEW

2.1 Conceptual Review

2.1.1 The Evolution of Farm Record Management

Agricultural data form the foundational basis of modern farm management. Historically, practitioners relied extensively on pen-and-paper methodologies, unstructured logbooks, and human memory to document operational inputs and crop yields. While these traditional, free-form methods possess historical provenance and flexibility, they severely restrict data mobility and are highly susceptible to fragmentation, physical loss, recall bias, and manual transcription errors. The transition toward digital recordkeeping initially introduced rudimentary spreadsheets; however, these lacked automated interoperability and were prone to manipulation.

To address these inefficiencies, Farm Management Information Systems (FMIS) emerged as centralised, planned architectures designed to collect, process, store, and disseminate agricultural data (Fountas et al., 2015). A critical evolutionary flaw, however, occurred in farm administration: the separation of Farm Financial Accounts (FFA) from operational bioprocess records. As Poppe, Vrolijk, and Bosloper (2023) observed, financial accounting and farm-management systems originated from the same historical note-taking practices but diverged into separate software ecosystems as they computerised. This separation forces farmers to duplicate data entry across disjointed systems, increasing administrative burdens and preventing the accurate correlation of agronomic inputs with true financial profitability.

2.1.2 Decision-Support Systems (DSS)

Within agricultural informatics, a Decision-Support System (DSS) does not make isolated operational decisions; rather, it processes heterogeneous metrics—such as weather patterns, soil conditions, market prices, and historical yields—through deterministic rules or statistical and machine-learning models to provide stakeholders with actionable, optimised alternatives (van Klompenburg et al., 2020). By converting complex biological and economic variables into understandable visualisations and predictive estimates, DSS platforms bridge the gap between static farm records and

proactive resource management. Recent work demonstrates that such systems can be deployed even for resource-constrained smallholders when the underlying predictive models are deliberately kept inexpensive and mobile-deployable (Olisah, Smith, Smith, Lawrence, & Ojukwu, 2024).

2.1.3 The Shift from Manual to Digital Agriculture (Agriculture 4.0)

The agricultural sector is currently undergoing a disruptive transformation termed “Agriculture 4.0” or digital agriculture. This paradigm shifts farming from intuition-based practices to data-driven precision by integrating the IoT, artificial intelligence, robotics, and big-data analytics. Agriculture 4.0 operates across distinct architectural layers: sensors and IoT devices perform data sensing and connectivity, cloud computing handles large-scale storage, and AI algorithms perform predictive analysis. Because central cloud computing often struggles with latency and high bandwidth requirements in rural areas, modern precision agriculture increasingly utilises Edge Computing, which processes data locally—closer to the farm—thereby reducing latency and mitigating the vulnerabilities of poor rural internet connectivity (Mhlanga, 2023).

2.2 Theoretical Framework

2.2.1 Technology Acceptance and Innovation Systems Theories

Understanding why agricultural practitioners reject or adopt digital tools requires robust theoretical grounding. The literature frequently employs the Diffusion of Innovations (DOI) theory and the Unified Theory of Acceptance and Use of Technology (UTAUT) to explain that adoption depends heavily on a farmer’s perception of a technology’s relative advantage, usability, and complexity. Empirical FMIS-adoption research operationalises these constructs, showing that education, advisory access, and credit access materially shift the adoption decision (Carrer et al., 2017). Innovation Systems Theory further emphasises that successful digital transformation requires robust collaboration among technology providers, agricultural practitioners, advisory services, and policymakers.

2.2.2 The Theory of Bounded Rationality

Traditional agricultural economics often relies on “rational choice” theory, assuming farmers always act to maximise strict financial utility. However, agent-based modelling of agricultural communities demonstrates that farmers operate under “bounded

rationality.” This theory suggests that humans make decisions with limited cognitive and computational capacities; therefore, they seek “satisficing” (satisfactory and sufficient) solutions rather than mathematically optimal ones. Consequently, an effective FMIS must reduce the cognitive load on the farmer by presenting synthesised, clear options rather than overwhelming them with raw, unprocessed big data—a design principle directly reflected in the AGRI-PROFIT decision-support layer.

2.2.3 Socio-Technical Systems Perspective

From a socio-technical perspective, digitalisation fundamentally alters organisational structures, power relations, and rural societal dynamics. Technology adoption does not occur in a vacuum; socio-economic characteristics—such as a farmer’s education, age, and existing infrastructural limitations—directly constrain the successful integration of digital platforms (Mhlanga, 2023). Thus, engineering a farm-management platform requires addressing both the technical architecture and the social realities of the target demographic.

2.3 The Nigerian Context: Precision Agriculture and Mobile Scaling

The deployment of digital agriculture in developing regions heavily relies on mobile-mediated scaling. Unlike developed nations where large-scale autonomous machinery drives Agriculture 4.0, mobile phones provide the most accessible and cost-effective leapfrog technology for Sub-Saharan Africa (Mhlanga, 2023). Empirical meta-analyses demonstrate that Mobile Phone-Mediated Scaling of Agricultural Technologies (MPSAT) increases the odds of farmers adopting new agricultural practices, particularly when mobile tools are combined with traditional extension services, which can raise adoption odds by approximately 17%. A longitudinal analysis of precision agriculture in Nigeria identifies three phases: the introductory phase (1990–2005), the experimental phase (2006–2015), and the integration and scale-up phase (2016–2023).

However, scaling digital platforms in Nigeria encounters profound infrastructural bottlenecks. Severe deficits in reliable electricity and digital connectivity in rural farming communities critically hinder the deployment of cloud-heavy agritech. To mitigate these realities, localised deployment strategies must prioritise mobile-first, low-bandwidth software integrated with off-grid or solar-powered infrastructure. Furthermore, national integration policies such as the Food and Agriculture Organization’s (FAO) Digital Village Initiative (DVI) in Nigeria champion centralised,

ecosystem co-creation platforms to eliminate middlemen and specifically attract youth involvement through digital innovation.

2.4 Empirical Review and Socio-Technical Barriers to Adoption

2.4.1 Review of Existing Systems

Existing literature thoroughly documents the evolution of FMIS and Enterprise Resource Planning (ERP) systems (Fountas et al., 2015). Currently, commercial ERP software effectively handles internal payroll, invoicing, and supply-chain logistics, but such systems predominantly target large-scale, highly capitalised commercial farms (Carrer et al., 2017). In contrast, smallholder and medium-scale farmers frequently rely on simplistic, unstructured digital methods. While these rudimentary tools offer flexibility, they lack automation, restrict interoperability, and frequently succumb to data corruption.

2.4.2 Socio-Technical Barriers to Adoption

Despite the theoretical benefits of digital agriculture, empirical studies highlight several socio-technical barriers that systematically hinder widespread adoption:

- **Interoperability and Data Standards:** A primary technical roadblock is the absence of standardised data-exchange protocols among heterogeneous agricultural devices and software. Because current platforms operate as isolated silos, farmers face difficulties transferring data between applications, which severely restricts scalability (Tummers et al., 2019).
- **Data Fragmentation and Administrative Redundancy:** Operational field records and Farm Financial Accounts (FFA) traditionally occupy separate workflows, forcing farmers to duplicate data entry across disjointed systems and creating analytical inefficiencies (Poppe et al., 2023).
- **User Complexity and Digital Illiteracy:** Complex interfaces and feature-overloaded software deter adoption. A pronounced digital divide persists; GSMA data indicate that 68% of rural Nigerians lack smartphone ownership, compounded by systemic digital illiteracy and high inflation (Jaiyeola, 2023; Mhlanga, 2023).

- **Infrastructural and Economic Constraints:** High initial capital costs, unstable electricity, and limited internet connectivity in rural areas render many existing commercial FMIS platforms impractical for resource-constrained smallholders (Mhlanga, 2023; Nirosha, 2024).

2.5 Data-Sharing and Stakeholder Trust

A critical, often neglected aspect of farm management is the macro-economic necessity of generating verifiable data to secure financing. Historically, the use of ICT could not inherently prevent bias or manipulation in self-reported farm data, leading to a profound lack of trust from agricultural banks, investors, and insurance providers (Gebresenbet et al., 2023).

To overcome this, modern architectures increasingly integrate blockchain technology as a secure data-governance mechanism (Huang, 2020). Blockchain utilises an immutable, distributed digital ledger to store yield data, mechanisation logs, and financial metrics. By locking operational and financial records into a tamper-proof framework, the system provides a constant flow of credible, unaltered data. Consequently, this cryptographically secured transparency allows financial institutions and supply-chain partners to deploy smart contracts, automate insurance pay-outs, and accurately assess agricultural credit risk, fundamentally bridging the gap between rural producers and formal credit networks (Huang, 2020). Within the scope of the present study, however, AGRI-PROFIT operationalises this trust requirement through a more immediately deployable mechanism: authenticated, access-controlled, tokenised standardised reporting. A per-farm data boundary restricts each enterprise's records to its owner, while read-only, time-limited views expose standardised profit/loss and yield summaries to invited banks or investors. Full distributed-ledger governance is identified as a future extension that can layer tamper-evidence onto this foundation once the core platform is validated.

2.6 Structuring the Decision-Support Prediction Model for AGRI-PROFIT

Building on the conceptual review of decision-support systems (Section 2.1.2), this section establishes the methodological structure that AGRI-PROFIT will follow to convert unified operational and financial records into predictive decision support. The structure is organised as a layered pipeline, deliberately progressing from lightweight deterministic logic to optional data-driven models so that the platform remains viable

on accessible mobile processors and under intermittent connectivity (van Klompenburg et al., 2020; Olisah et al., 2024).

2.6.1 Data Layer and Feature Definition

The prediction pipeline begins with the unified data layer that distinguishes AGRI-PROFIT from siloed systems. Operational records (planting and harvest dates, input applications, labour hours, mechanisation usage, and observed yields) and financial records (input costs, sales revenue, and overheads) are automatically and atomically paired at the point of entry—each operation linked to its corresponding financial transaction through a shared activity category—rather than maintained as isolated ledgers. This pairing, enforced at the write layer, is what allows operational activity and its financial consequence to be analysed together. From these records, candidate predictor variables (features) are defined: agronomic features (crop type, area cultivated, input quantities, season), resource features (machine-hours, labour-hours, fuel), and contextual features (location, rainfall band, soil class where available). The corresponding prediction targets are crop yield, gross margin or profit/loss, and recommended input allocation.

2.6.2 Data Pre-processing

Because smallholder records are typically sparse and inconsistent, a pre-processing stage is essential before any model is trained or applied. This stage handles missing values through imputation, harmonises units and date intervals, removes obvious outliers, and normalises numeric ranges. Olisah et al. (2024) demonstrate that a carefully designed pre-processing pipeline—addressing inconsistent year intervals, missing data, and variables inaccessible to smallholders—is decisive for prediction quality in Sub-Saharan farming contexts. AGRI-PROFIT adopts this as a design principle for real-farm data; in the current release, in which the forecast tier is trained on clean synthetic data, pre-processing is limited to categorical encoding and passthrough of numeric features, with the fuller pipeline described here applying as farms accumulate real records.

2.6.3 A Tiered Modelling Strategy

Rather than committing to a single algorithm, AGRI-PROFIT adopts a tiered modelling strategy that aligns model complexity with available data and device capability:

- **Tier 1 – Deterministic, rule-based engine (baseline).** Transparent IF–THEN agronomic and financial rules compute gross margin, break-even points, and input thresholds directly from the records. This tier runs fully offline, requires no training data, and is explainable—satisfying the bounded-rationality requirement for clear, satisficing recommendations (Section 2.2.2).
- **Tier 2 – Statistical regression (interpretable prediction).** Once sufficient historical records accumulate, multiple linear or regularised regression estimates yield and profitability as functions of the defined features, quantifying the contribution of each input to the outcome.
- **Tier 3 – Lightweight machine learning (optional, server-assisted).** Where data volume permits, tree-based ensembles such as Random Forest and Extreme Gradient Boosting (XGBoost) are recommended, as they consistently achieve strong accuracy on tabular agricultural data and are far less resource-intensive than deep neural networks (van Klompenburg et al., 2020; Olisah et al., 2024).

This progression describes a general methodological ladder drawn from the literature. The AGRI-PROFIT platform realises two of its rungs as its implemented two-tier design: the deterministic rule-based tier (Tier 1) and the tree-based Random Forest tier (Tier 3), with the intermediate regression tier (Tier 2) available as an optional future refinement. This ensures the platform delivers value from day one through the deterministic tier, while providing a clear, evidence-based pathway to predictive analytics as the user’s data matures.

2.6.4 Model Training, Validation, and Evaluation Metrics

For the statistical and machine-learning tiers, data are partitioned into training and testing sets using a hold-out split, guarding against over-fitting on small datasets (van Klompenburg et al., 2020). In the current implementation, continuous predictions (yield) are evaluated using the coefficient of determination (R^2) and the Mean Absolute Error (MAE); the Root Mean Square Error (RMSE) is a standard complementary measure for such tasks and may additionally be reported, and Olisah et al. (2024) propose combining RMSE and MAE into a single robustness measure. For the rule-based tier, evaluation compares system-computed margins against farmers’ actual recorded outcomes to confirm arithmetic fidelity.

2.6.5 From Prediction to Decision Support

The final stage translates predicted values into recommendations. Rather than presenting raw numbers, AGRI-PROFIT converts predictions into ranked, actionable options—for example, the input mix that maximises expected gross margin, or an early warning when projected costs approach the break-even threshold. This deliberately reduces cognitive load and supports satisficing decisions (Section 2.2.2), and the structured, standardised outputs simultaneously feed the trusted data-sharing layer described in Section 2.5 (Huang, 2020). Deployment remains mobile-first and low-bandwidth: the deterministic tier requires no training data and runs on modest server hosting, with results delivered to the mobile client, while the optional forecast tier likewise runs server-side and its outputs are synchronised when connectivity is available (Mhlanga, 2023).

2.7 Gap in Literature

The systematic review of the literature reveals a pronounced bias in the design and deployment of existing FMIS and digital agricultural tools. Most commercially available platforms cater to the operational scale, financial capital, and robust broadband infrastructure typical of developed nations (Carrer et al., 2017; Fountas et al., 2015). Consequently, there is a severe lack of lightweight, localised frameworks tailored specifically for the socio-technical and infrastructural realities of Nigerian agriculture, which remains dominated by smallholders and medium-scale enterprises facing distinct infrastructural deficits (Mhlanga, 2023).

Furthermore, while studies acknowledge the chronic problem of data fragmentation between agricultural operations and financial accounting (Pope et al., 2023), there is a critical scarcity of conceptual system architectures that successfully integrate these disparate data streams within a low-bandwidth, mobile-first platform and convert them into appropriately lightweight predictive decision support (van Klompenburg et al., 2020; Olisah et al., 2024). Finally, current tools focus almost exclusively on internal farm management, neglecting the macro-economic necessity of secure data-sharing protocols—such as blockchain—to build operational trust with agricultural banks and investors (Huang, 2020). The AGRI-PROFIT platform directly addresses these identified gaps by engineering a unified, socio-technically appropriate system architecture for the Nigerian agricultural sector.

CHAPTER THREE

SYSTEM DESIGN AND METHODOLOGY

3.1 Introduction

This chapter presents the design and development methodology adopted for the AGRI-PROFIT platform and the methods used to evaluate it against the aim and objectives established in Chapter One. It describes the research and development approach, the requirements analysis, the overall system architecture, the unified data model, the design of each functional module, the security design, and the evaluation methodology. The chapter is written to describe the target architecture of the platform—that is, the complete system the design specifies—while stating explicitly, where relevant, which capabilities constitute the current release and which are deferred as future work. This distinction is maintained throughout so that the methodology remains an honest account of the system as designed and built.

The four objectives of the study are addressed by distinct but connected methods. Objective 1 (identifying socio-technical barriers) is addressed through the requirements analysis and through the inclusive design of low-barrier data-entry channels. Objective 2 (translating records into decision-support metrics) is addressed through the unified data model and the deterministic decision-support tier. Objective 3 (secure, trusted data-sharing) is addressed through the authentication and per-farm boundary design and the tokenised stakeholder-reporting mechanism. Objective 4 (evaluation of usability, cost-effectiveness, and viability) is addressed through the multi-strand evaluation methodology presented in Section 3.8.

3.2 Research and Development Methodology

The platform was developed using an iterative, incremental methodology consistent with Rapid Application Development (RAD) principles, which are well suited to systems whose requirements are refined through successive prototypes rather than fixed entirely in advance. Development proceeded in short cycles, each producing a working increment that was verified before the next was undertaken. Version control (Git) was used throughout, with each functional increment committed as a discrete, reversible

checkpoint, allowing safe experimentation and providing a traceable development history.

Development was assisted by a command-line AI coding agent operating within the project repository, guided by a curated set of engineering skills. The agent produced code increments against explicit, human-reviewed specifications; each increment was inspected, tested, and verified by the researcher before acceptance. This human-in-the-loop arrangement preserved the researcher’s authorship of design decisions while accelerating implementation. Two governing documents—a domain context glossary and a product requirements document—were maintained under version control to keep terminology, scope, and implementation consistent across the codebase and the written work.

3.3 System Requirements Analysis

Requirements were derived from the problem analysis in Chapters One and Two and organised into functional and non-functional categories. The non-functional requirements are of particular importance because they encode the socio-technical constraints that this study exists to address (Objective 1).

3.3.1 Functional Requirements

The platform is required to: record farm operations and their associated costs as paired entries; compute profit-and-loss and gross-margin metrics from those records; track equipment, maintenance, and depreciation; produce deterministic decision-support metrics (unit cost of production and per-crop gross margin) from real records; provide an optional yield forecast; authenticate users and confine each user’s data to their own farm; and allow a farmer to grant a bank or investor a read-only view of standardised reports.

3.3.2 Non-Functional Requirements

- **Low-bandwidth, mobile-first operation:** the interface must be usable on affordable smartphones over constrained networks.
- **Offline data capture:** operations must be recordable without connectivity and synchronised, without loss or duplication, once a connection returns.

- **Security and data isolation:** access must be authenticated and each farm's data isolated from every other.
- **Accessibility for low-specification devices and feature phones:** data entry must be possible through channels that do not presuppose a smartphone or reliable data connection.
- **Computational feasibility:** decision-support processing must run within the resources of accessible mobile processors and modest server hosting.

3.4 System Architecture

3.4.1 Overall Architecture

AGRI-PROFIT adopts a client-server architecture comprising a mobile-first Progressive Web Application (PWA) client and a backend application-programming-interface (API) service. The PWA client is installable on a smartphone and is responsible for data entry, local queuing of records while offline, and presentation of reports and decision-support outputs. The backend exposes a thin HTTP interface over a layered structure in which business logic and database access are separated from the transport layer, promoting testability and maintainability. Data are persisted in a relational database whose schema is managed through versioned migrations.

3.4.2 Technology Stack

The client is implemented with a modern component-based JavaScript framework compiled to a lightweight bundle, with a local browser database used to queue records for offline-first capture. The backend is implemented in Python using an asynchronous web framework with an object-relational mapper and a migration tool for schema evolution. These choices favour a small deployment footprint suitable for modest, low-cost hosting, in keeping with the cost-effectiveness objective.

3.5 Data Model Design

The core design decision of AGRI-PROFIT is the elimination of the historical separation between operational records and financial accounts. Rather than maintaining two independent ledgers, the data model pairs each operation with its financial consequence at the point of entry. An Operational Log (a timestamped record of a physical farm activity) is automatically and atomically paired with a corresponding

Financial Transaction, the two being linked through a shared Activity Category. This pairing is enforced at the write layer of the application, so that in normal operation every recorded activity carries its financial effect with it.

The financial ledger is single-entry: each Financial Transaction is classified as either an expense (debit) or revenue (credit) and tagged with an Activity Category, without an enforced double-entry debit/credit pairing. This is a deliberate simplification appropriate to the smallholder context and to the analytical purpose of the platform, and it is stated as such to avoid over-representing the accounting model. The paired-record design is what allows operational activity and its financial outcome to be analysed together, and it is the structural foundation on which the decision-support tier depends.

The principal entities of the data model are summarised in Table 3.1.

Table 3.1: Principal entities of the AGRI-PROFIT data model

Entity	Role in the model
Operational Log	Timestamped physical farm activity; the primary unit of operational entry.
Financial Transaction	Expense or revenue record paired with an Operational Log via a shared Activity Category.
Equipment	Machinery tracked for acquisition cost, depreciation, and maintenance.
Maintenance Log	A service or repair event recorded against a specific Equipment item.
Farm / User	Ownership boundary; scopes all records to a single enterprise (security design).

3.6 Module Design

The platform is organised into cohesive modules, each with a defined responsibility. The design of each is described below; where a module is partially realised in the current release, the boundary between built and deferred behaviour is stated explicitly.

3.6.1 Offline Operational Logbook

This module handles data entry for operations, yields, and post-harvest bioprocesses. To satisfy the offline-capture requirement, records entered without connectivity are

queued in a local browser database and synchronised to the backend when a connection is restored. Synchronisation is idempotent: each queued record carries a client-generated identifier, so that replaying the same record—for example after a connection is lost mid-entry—results in a single stored record rather than a duplicate. In the current release, offline support is provided for the data-entry (write) path; retrieval of previously stored data assumes connectivity unless runtime caching of read endpoints is enabled, which is identified as a planned enhancement.

3.6.2 Financial Ledger Engine

This module is the translation layer that creates the paired Financial Transaction for each Operational Log, applying the shared Activity Category and an optional tax label. Its correctness is central to the platform, since all downstream reporting and decision support derive from the ledger it maintains.

3.6.3 Reporting

The reporting module aggregates the ledger into profit-and-loss statements—overall, per-category, and monthly—and produces standardised, exportable reports. These exports provide the standardised-reporting basis on which the stakeholder-sharing mechanism (Section 3.6.7) is built.

3.6.4 Mechanisation and Equipment Tracker

This module records equipment acquisition cost, maintenance events, and depreciation, and records mechanisation usage and fuel as operational entries tagged with the mechanisation activity category, so that machinery costs are captured within the ledger alongside other activities (addressing the mechanisation-tracking element of the aim). A direct per-machine association between an individual operation and a specific equipment item is identified as a future refinement.

3.6.5 Predictive Decision-Support Engine

The decision-support engine follows a two-tier design that deliberately aligns model complexity with available data and device capability (van Klompenburg, Kassahun, & Catal, 2020; Olisah, Smith, Smith, Lawrence, & Ojukwu, 2024).

- **Tier 1 – Deterministic rule-based tier.** Transparent rules compute unit cost of production and per-crop gross margin directly from the farm’s real

Operational Logs and Financial Transactions, grouped by Activity Category and crop. This tier requires no training data, runs without training data on modest server hosting, and is fully explainable, providing clear, satisficing decision support consistent with the bounded-rationality principle discussed in Chapter Two.

- **Tier 2 – Statistical yield forecast (optional).** A Random Forest regressor produces a yield estimate from agronomic inputs such as rainfall, fertiliser, and soil pH, reporting error measures suited to regression tasks.

An explicit disclosure governs the forecast tier: in the current release, the Random Forest model is trained on representative synthetic data rather than on any individual farm's own records, and it does not learn or adapt from farm usage. This design choice reflects the reality that a newly deployed farm has insufficient historical data to train a reliable model; retraining on real accumulated records is identified as future work. The deterministic tier, by contrast, operates entirely on the farm's real records. This distinction is stated plainly to ensure the platform's analytical claims are not overstated.

For the statistical tier, model quality is expressed in the current implementation using the coefficient of determination (R^2) and the Mean Absolute Error (MAE); the Root Mean Square Error (RMSE) is a standard complementary measure for continuous prediction tasks in agricultural yield modelling and may additionally be reported (van Klompenburg et al., 2020; Olisah et al., 2024).

3.6.6 Authentication and Per-Farm Data Boundary

To satisfy the security requirement embedded in the aim and in Objective 3, the platform authenticates users and confines each user's records to their own farm. Authentication issues a signed token upon login, with passwords stored only in hashed form. Every read and write operation across the ledger, reporting, equipment, and decision-support paths is scoped to the authenticated user's farm, so that one enterprise's data cannot be reached by another. The farm/user boundary is introduced through a versioned database migration rather than ad-hoc schema creation, preserving the integrity of the managed schema.

3.6.7 Tokenised Stakeholder Reporting

The trust-building element of Objective 3 is realised through a tokenised, read-only sharing mechanism. A farm owner may mint a revocable, opaque token that grants a bank or investor read-only access to that farm’s standardised profit-and-loss and yield summaries, without the recipient holding an account. The token cannot be used to write data or to reach any other farm, and it can be revoked at will. Within the scope of this study, this authenticated, access-controlled, tokenised reporting is the mechanism by which standardised data is shared with financial stakeholders; immutable distributed-ledger (blockchain) governance, reviewed in Chapter Two, is designed as a future extension that can layer tamper-evidence onto this foundation.

3.6.8 Feature-Phone (USSD/SMS) Data Entry

To address the socio-technical barrier at the heart of Objective 1—that a large proportion of rural farmers lack smartphones—the design includes an inbound message channel through which a farm activity can be recorded from a feature phone. An inbound webhook accepts a message consisting of a sender identifier and a short structured text, parses it into an activity and amount, and creates a paired Operational Log and Financial Transaction through the same ledger service used by the application, so that a phone-entered record is indistinguishable from an app-entered one and appears in the same reports. The sender’s number is mapped to a registered farm so that messages are routed to the correct ledger and unregistered senders are rejected. The channel is designed to be gateway-agnostic, so that it can be exercised through simulated inbound-message payloads independently of any carrier; implementation and integration with a live carrier gateway are identified as future work. This keeps the contribution—an inclusive, low-barrier entry path—demonstrable without dependence on a commercial messaging service.

3.7 Security Design

Security is treated as a first-class requirement because the word “secure” appears in the aim and because Objective 3 depends on stakeholders trusting the platform’s outputs. The security design rests on three elements: authenticated access with hashed credentials; strict per-farm data isolation enforced on every data path; and scoped, revocable, read-only tokens for external sharing that cannot escalate to write access or cross farm boundaries. Together these ensure that data sharing is deliberate and

controlled rather than open, which is the precondition for the trust that Objective 3 seeks to establish.

3.8 Evaluation Methodology

Objective 4 requires an empirical evaluation of the platform’s usability, cost-effectiveness, and operational viability under Nigerian constraints. A full field deployment with a large sample of smallholder farmers represents the ideal evaluation; however, such a deployment fell outside the time and access constraints of this study and is therefore presented as designed future work (Section 3.8.5). The evaluation actually undertaken comprises four complementary strands, each producing evidence that can be reported in the results chapter. The strands are chosen so that meaningful, honest evidence of viability is obtained without a field study, while the more extensive evaluation is fully specified for future execution.

3.8.1 Functional Verification

Each functional requirement is verified against defined test cases using an automated test suite, supplemented by manual walkthroughs of the core workflows. For the deterministic decision-support tier, verification asserts that the computed unit cost of production and gross margin match values calculated by hand from seeded ledger records, confirming arithmetic fidelity. For the offline path, a test confirms that a queued record synchronises exactly once and that replaying the same client identifier yields a single stored record. The outcome of this strand is a requirement-by-requirement statement of pass or fail with the corresponding test evidence.

3.8.2 Performance Benchmarking

Performance under constrained conditions is measured using an automated auditing tool (Lighthouse) run against the production build of the application, with device emulation set to a mobile profile and network throttling applied to approximate a slow mobile connection. The reported measures are the overall performance score together with load-timing metrics—First Contentful Paint, Largest Contentful Paint, Total Blocking Time, and Speed Index—and the transferred bundle size. To reduce noise, the audit is run several times and the median reported. This strand directly evidences the low-bandwidth claim, and any performance limitation it reveals (for example, the cost of charting libraries) is reported as a concrete finding rather than concealed.

3.8.3 Hosting-Cost Analysis

Cost-effectiveness is evaluated through a desk-based analysis of the projected monthly cost of hosting the platform on a modest cloud or virtual-private-server tier. The analysis estimates the platform's resource footprint and applies published provider pricing to derive an indicative monthly operating cost, clearly labelled as an estimate. This provides a defensible figure for the viability discussion without requiring a paid production deployment.

3.8.4 Usability Evaluation (Heuristic and SUS)

Usability is evaluated through an expert heuristic evaluation combined with a standardised usability questionnaire, a recognised and citable approach that is appropriate where a large end-user sample is not available (Nielsen & Molich, 1990; Nielsen, 1994). Three to five evaluators—comprising the project supervisor and peers—independently perform a fixed set of core tasks (recording an operation with its cost, viewing the profit-and-loss report, generating a yield forecast, and exporting or sharing a report) and then assess the interface against Nielsen's ten usability heuristics, recording issues with severity ratings. Each evaluator additionally completes the ten-item System Usability Scale (SUS), which yields a single score between 0 and 100 (Brooke, 1996). With a small evaluator panel the SUS score is reported as indicative rather than statistically powered. The instruments used for this strand—the task script, heuristic checklist, and SUS questionnaire—are provided as a separate evaluation pack. The outputs of this strand are a prioritised list of usability issues with severity ratings and an indicative SUS score.

3.8.5 Designed Field Evaluation (Future Work)

For completeness and to demonstrate methodological awareness, the ideal field evaluation is specified even though it is not executed in this study. It would recruit a representative sample of smallholder and medium-scale farmers, obtain informed consent, provide a short onboarding session, and have participants use the platform for their own record-keeping over a defined period. It would capture task-completion rates, error rates, SUS scores across a larger sample, and qualitative feedback, and would measure real device performance and data usage in the field. This design is presented as the recommended next step beyond the present work.

3.9 Summary

This chapter has set out the methodology by which AGRI-PROFIT was designed, developed, and evaluated. It described an iterative, version-controlled development approach with human-reviewed, agent-assisted implementation; a requirements analysis grounded in the socio-technical constraints of Nigerian agriculture; a client–server architecture built around a mobile-first offline-capable client; and a unified data model in which each operation is atomically paired with its financial consequence. It specified the design of each module, including a two-tier decision-support engine whose deterministic tier operates on real records and whose optional forecast tier is transparently trained on synthetic data pending sufficient real history. Finally, it presented a four-strand evaluation methodology—functional verification, performance benchmarking, hosting-cost analysis, and heuristic-plus-SUS usability evaluation—together with a fully specified field evaluation designed as future work. The results of applying this methodology are presented in Chapter Four.

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[Note to author — remove before submission: the following in-text citations still need full, verified APA entries added above: Abbasi, Martinez, & Ahmad (2022); Abiri et al. (2023); Basir et al. (2024); Dayioglu & Turker (2021); Gebresenbet et al. (2023); Giua, Materia, & Camanzi (2020); Husemann & Novkovic (2012); Jaiyeola (2023); Javaid et al. (2022); Nirosha (2024); Poppe, Vrolijk, & Bosloper (2023); Tummers, Kassahun, & Tekinerdogan (2019).]